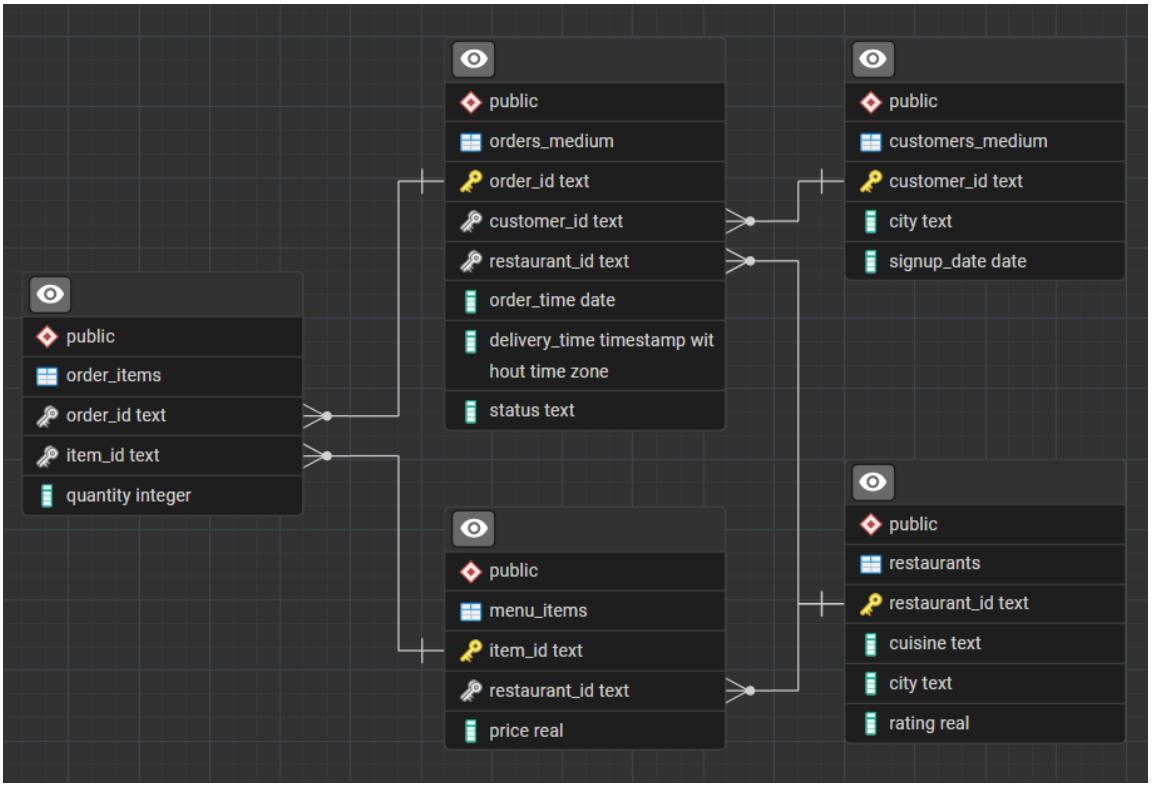
Food Delivery: Queries & Insights

Queries

-- ERD



-- Q1 Number of customers in each city

```
SELECT city, COUNT (customer_id)
FROM customers_medium
GROUP BY 1
ORDER BY 2 DESC;
```

	city text	count bigint
1	Bristol	270
2	Leeds	252
3	Liverpool	247
4	London	246
5	Birmingham	245
6	Manchester	240

-- Q2 Number of restaurants under each cuisine

```sql
SELECT cuisine, COUNT(restaurant_id)
FROM restaurants
GROUP BY 1
ORDER BY 2 DESC;
```

	cuisine text	count bigint
1	Thai	26
2	Indian	22
3	Italian	20
4	Mexican	20
5	American	19
6	Chinese	13

-- Q3 Top rated restaurants

```sql
SELECT
    restaurant_id, rating,
    CASE
        WHEN rating = 5 THEN 'Excellent'
        WHEN rating >= 4 THEN 'Good'
        WHEN rating >= 3 THEN 'Average'
        ELSE 'Poor'
    END AS rating_category
FROM restaurants
ORDER BY 2 DESC;
```

	restaurant_id [PK] text	rating real	rating_category text
1	R020	5	Excellent
2	R030	5	Excellent
3	R115	5	Excellent
4	R093	4.9	Good
5	R042	4.9	Good
6	R083	4.9	Good
7	R086	4.9	Good

-- Q4 Top 5 Restaurants with the highest number of customers

```
SELECT
    r.restaurant_id,
    COUNT(DISTINCT o.customer_id) AS num_of_customers
FROM restaurants r
LEFT JOIN orders_medium o
ON r.restaurant_id = o.restaurant_id
GROUP BY r.restaurant_id
ORDER BY num_of_customers DESC
LIMIT 5
```

	restaurant_id [PK] text	num_of_customers bigint
1	R021	57
2	R060	55
3	R096	54
4	R038	53
5	R046	53

-- Q5 Average delivery time per restaurant

```
SELECT
    r.restaurant_id,
    ROUND (AVG(EXTRACT(EPOCH FROM (o.delivery_time -
o.order_time)) / 60), 2) AS avg_delivery_minutes
FROM orders_medium o
JOIN restaurants r
ON o.restaurant_id = r.restaurant_id
GROUP BY r.restaurant_id
ORDER BY avg_delivery_minutes;
```

	restaurant_id [PK] text	avg_delivery_minutes numeric
1	R010	45.98
2	R115	47.12
3	R084	48.23
4	R044	48.53
5	R120	49.53
6	R018	49.64
7	R111	49.72
8	R116	49.81
9	R015	50.00
10	R095	50.34

-- Q6 Top-Grossing Restaurants by Cuisine

```
SELECT
    r.restaurant_id, r.cuisine,
    ROUND(SUM(i.quantity * m.price)::numeric, 2) AS revenue
FROM restaurants r
JOIN orders_medium o
ON r.restaurant_id = o.restaurant_id
JOIN order_items i
ON o.order_id = i.order_id
JOIN menu_items m
ON i.item_id = m.item_id
GROUP BY r.restaurant_id, r.cuisine
ORDER BY revenue DESC;
```

	restaurant_id [PK] text	cuisine text	revenue numeric
1	R061	Thai	6232.12
2	R021	Italian	6108.38
3	R032	American	5671.31
4	R011	Italian	5664.40
5	R046	Mexican	5596.77
6	R060	Indian	5364.45
7	R067	Italian	5260.59
8	R088	American	5250.62
9	R055	American	5141.67
10	R026	Indian	5137.67

-- Q7 Ranking Restaurants by Orders

```
SELECT
    restaurant_id,
    COUNT(*) AS total_orders,
    DENSE_RANK() OVER (ORDER BY COUNT(*) DESC) AS rank
FROM orders_medium
GROUP BY restaurant_id;
```

	restaurant_id text	total_orders bigint	rank bigint
1	R021	57	1
2	R060	56	2
3	R022	55	3
4	R038	55	3
5	R088	54	4
6	R096	54	4
7	R046	53	5
8	R032	53	5
9	R074	53	5
10	R061	51	6
11	R078	51	6

-- Q8 Customer Orders and Spending Summary

```
SELECT
    c.customer_id,
    ROUND(COALESCE(SUM(o1.quantity * m.price), 0)::numeric, 2) AS
total_spent,
    ROUND(
        CASE
            WHEN COUNT(DISTINCT o.order_id) = 0 THEN 0
            ELSE SUM(o1.quantity * m.price)::numeric /
COUNT(DISTINCT o.order_id)
        END,
        2
    ) AS avg_per_order,
    COUNT(DISTINCT o.order_id) AS num_of_orders
FROM customers_medium c
LEFT JOIN orders_medium o
    ON c.customer_id = o.customer_id
LEFT JOIN order_items o1
    ON o.order_id = o1.order_id
LEFT JOIN menu_items m
    ON m.item_id = o1.item_id
GROUP BY c.customer_id;
ORDER BY total_spent DESC;
```

	customer_id [PK] text	total_spent numeric	avg_per_order numeric	num_of_orders bigint
1	C0001	606.47	151.62	4
2	C0002	422.18	84.44	5
3	C0003	454.91	90.98	5
4	C0004	233.12	233.12	1
5	C0005	231.84	77.28	3
6	C0006	570.65	95.11	6
7	C0007	910.67	113.83	8
8	C0008	144.29	48.10	3
9	C0009	0.00	0.00	0
10	C0010	134.60	67.30	2
11	C0011	521.80	86.97	6
12	C0012	230.22	76.74	3
13	C0013	153.68	76.84	2

-- Q9 Top-Selling Menu Item per Restaurant

```sql
SELECT
    restaurant_id,
    item_id AS top_item,
    total_quantity_ordered
FROM (
    SELECT
        m.restaurant_id,
        m.item_id,
        SUM(o.quantity) AS total_quantity_ordered,
        RANK() OVER (
            PARTITION BY m.restaurant_id
            ORDER BY SUM(o.quantity) DESC
        ) AS rank
    FROM menu_items m
    JOIN order_items o
    ON m.item_id = o.item_id
    GROUP BY m.item_id, m.restaurant_id
) sub
WHERE rank = 1
ORDER BY restaurant_id;
```

	restaurant_id text	top_item text	total_quantity_ordered bigint
1	R001	M0265	53
2	R002	M0143	64
3	R003	M0005	47
4	R004	M0305	55
5	R005	M0072	51
6	R006	M0247	68
7	R007	M0135	63
8	R008	M0270	65
9	R009	M0223	66
10	R010	M0279	55

-- Q10 The revenue generated each month in each year

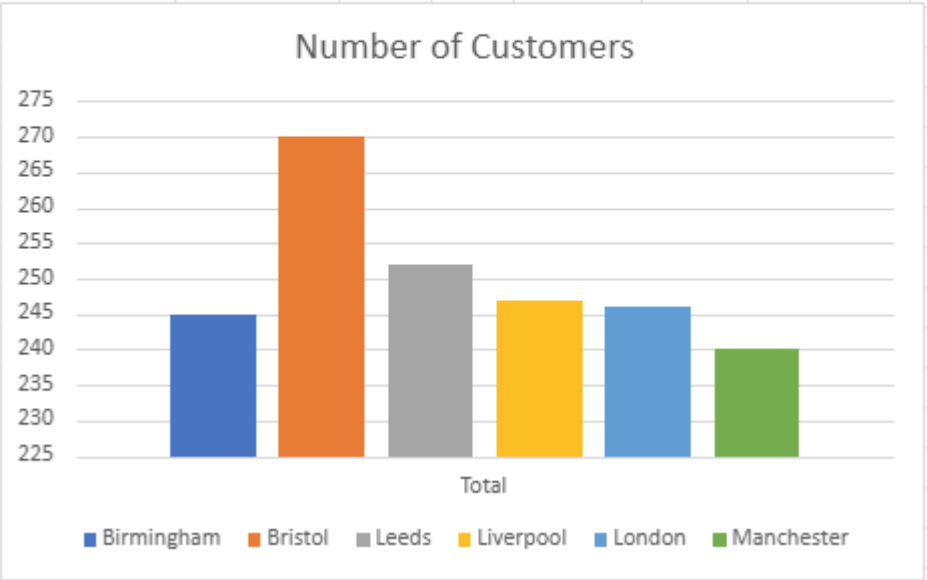
```
SELECT
    TO_CHAR(signup_date, 'MM') AS signup_month,
    COUNT(customer_id) AS num_customers,
    COUNT(customer_id) - LAG(COUNT(customer_id)) OVER (ORDER BY
TO_CHAR(signup_date, 'MM')) AS diff_from_prev_month
FROM customers_medium
GROUP BY signup_month
ORDER BY signup_month;
```

	signup_month text	num_customers bigint	diff_from_prev_month bigint
1	01	110	[null]
2	02	110	0
3	03	146	36
4	04	141	-5
5	05	127	-14
6	06	112	-15
7	07	138	26
8	08	131	-7
9	09	136	5
10	10	142	6
11	11	150	8
12	12	57	-93

Insights

Query 1:

	city					
	Birmingham	Bristol	Leeds	Liverpool	London	Manchester
Sum of count	245	270	252	247	246	240



Observations: Orders are pretty evenly spread across the top cities, with Bristol just taking the lead, followed closely by Leeds and Liverpool. London, Birmingham, and Manchester aren't far behind, showing that customer demand is strong and consistent across all these major cities, rather than being dominated by just one.

Query 2:

cuisine	Sum of count
American	19
Chinese	13
Indian	22
Italian	20
Mexican	20
Thai	26
Grand Total	120

Observations: Thai cuisine comes out on top, with Indian and Italian close behind. Mexican and American options show similar popularity, while Chinese cuisine has a slightly lower count. Overall, preferences are quite varied, with no single cuisine dominating too strongly.

Query 3:

	rating_category			
	Excellent	Average	Good	Grand Total
Count of restaurant_id	3	56	61	120

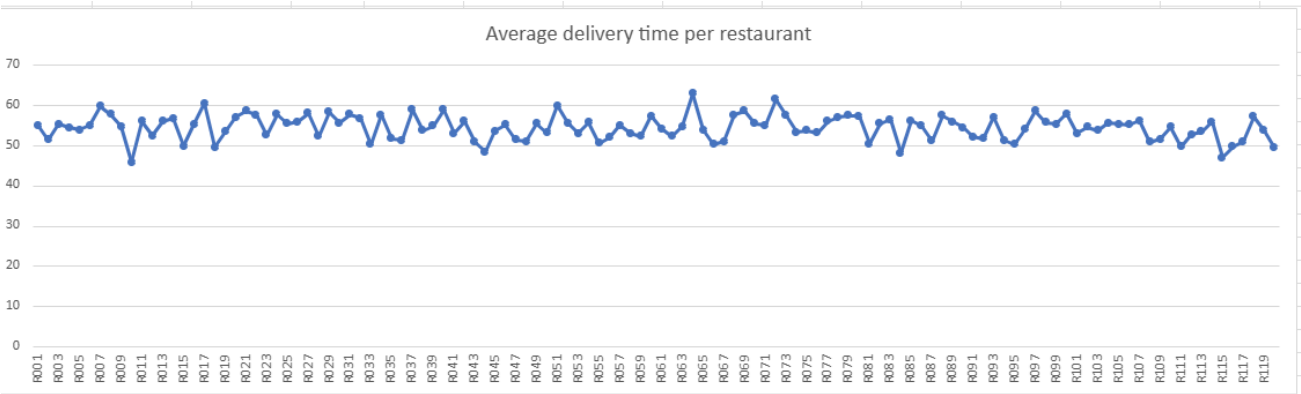
Observations: Most restaurants are rated quite highly, with a large number falling into the 'Good' category, typically between 4.0 and 4.9. Only a few restaurants achieve a perfect 'Excellent' rating, while ratings in the 'Average' range are more spread out and generally lower. Overall, this suggests that customer satisfaction is strong across the majority of restaurants.

Query 4:

	restaurant_id				
	R046	R038	R096	R060	R021
Sum of num_of_customers	53	53	54	55	57

Observations: The top five restaurants have very similar customer numbers, with R021 slightly ahead at 57 customers. R060 and R096 are close behind, while R038 and R046 follow with only a small difference. Overall, customer visits are fairly evenly spread across these leading restaurants.

Query 5:



Observations: A few restaurants stand out at both ends of the spectrum. On the faster side, restaurants like R010 and R115 have noticeably lower average delivery times, suggesting more efficient operations or better logistics. On the slower end, restaurants such as R064, R072, and R017 take significantly longer to deliver, which could be due to higher demand, longer preparation times, or delivery challenges. These differences highlight opportunities for improvement as well as examples of best-performing restaurants.

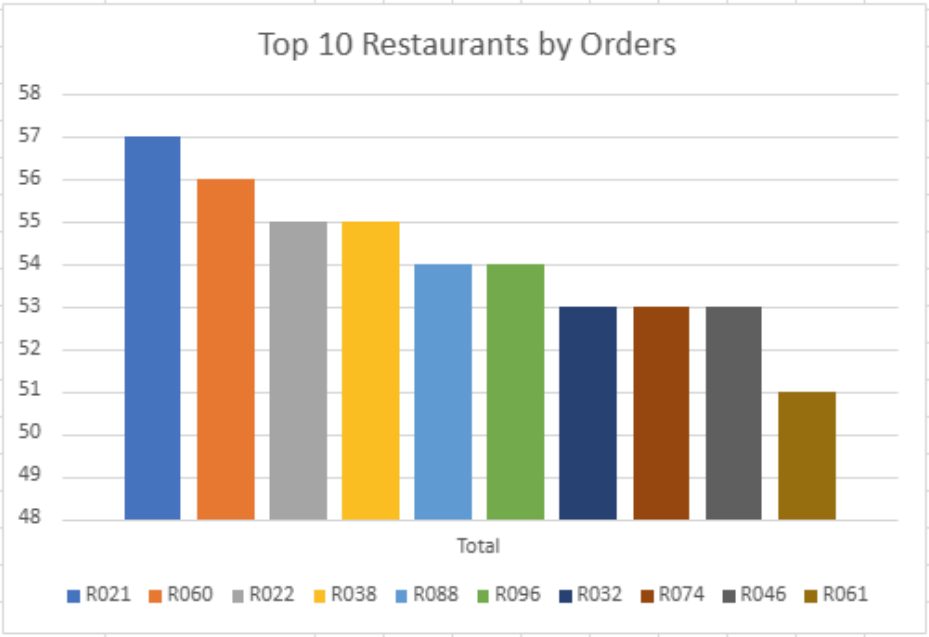
Query 6:

restaurant_id	Cuisine	Revenue
R061	Thai	6232.12
R021	Italian	6108.38
R032	American	5671.31
R011	Italian	5664.4
R046	Mexican	5596.77
R060	Indian	5364.45
R067	Italian	5260.59
R088	American	5250.62
R055	American	5141.67

Observations: The top 10 restaurants by revenue show that Italian cuisine performs the best, with three restaurants (R021, R011, and R067) among the top earners. Thai, American, Mexican, and Indian cuisines also appear in the list, but Italian leads in overall gross. The highest-grossing restaurant is R061 serving Thai cuisine, closely followed by Italian restaurants, indicating strong customer preference and sales for these cuisines.

Query 7:

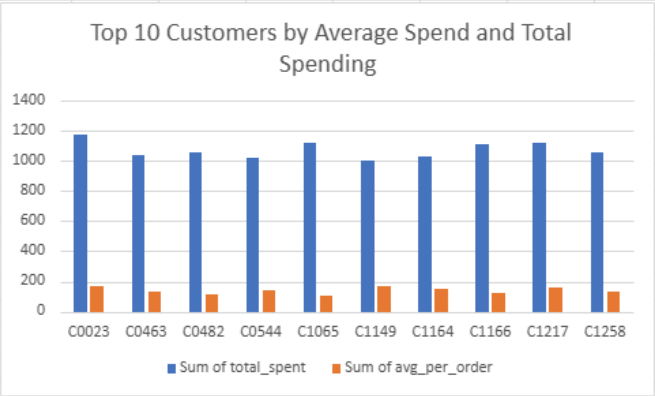
	restaurant_id									
	R021	R060	R022	R038	R088	R096	R032	R074	R046	R061
Sum of total_orders	57	56	55	55	54	54	53	53	53	51



Observations: Looking at the top restaurants by total orders, R021 takes the lead with 57 orders, just ahead of R060 with 56. R022 and R038 share third place with 55 orders, while R088 and R096 aren't far behind with 54. R046, R032, and R074 round out the top spots with 53 orders each. Overall, the top restaurants are all doing really well, with order numbers pretty close across the board.

Query 8:

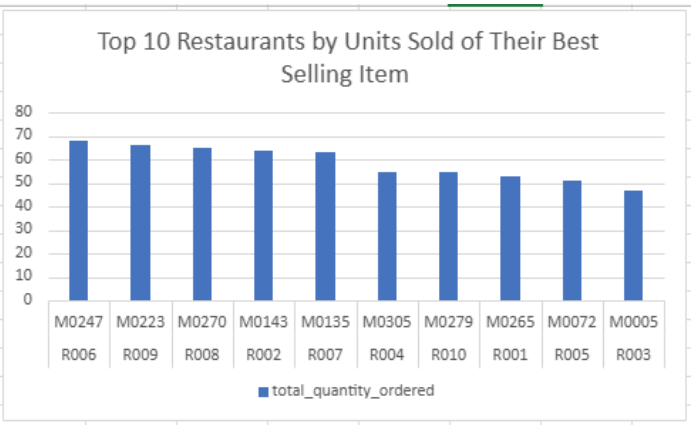
customer_id	Sum of total_spent	Sum of avg_per_order
C0023	1173.1	167.59
C0463	1039.87	129.98
C0482	1055.17	117.24
C0544	1019.38	145.63
C1065	1122.06	102.01
C1149	998.83	166.47
C1164	1026.12	146.59
C1166	1108.56	123.17
C1217	1121.95	160.28
C1258	1056.1	132.01



Observations: Looking at the top customers by total spending, C0023 leads with $1,173.10 spent over 7 orders, averaging $167.59 per order. Other high spenders, like C1065 and C1217, also show strong totals, with C1065 making more frequent but slightly smaller purchases, while C1217 has higher average spend per order. Overall, these customers demonstrate a mix of loyal, frequent buyers and those who spend more per visit, highlighting key contributors to revenue.

Query 9:

restaurant_id	top_item	total_quantity_ordered
R006	M0247	68
R009	M0223	66
R008	M0270	65
R002	M0143	64
R007	M0135	63
R004	M0305	55
R010	M0279	55
R001	M0265	53
R005	M0072	51
R003	M0005	47



Observations: Looking at each restaurant's top-selling item, R006 leads with 68 orders of M0247, closely followed by R009 with 66 orders of M0223 and R008 with 65 of M0270. Most restaurants have their top item ordered between 50–65 times, showing that while popular items vary, each restaurant has a clear favorite that drives significant sales.

Query 10:

signup_month	num_customers	diff_from_prev_month
1	110	NULL
2	110	0
3	146	36
4	141	-5
5	127	-14
6	112	-15
7	138	26
8	131	-7
9	136	5
10	142	6
11	150	8
12	57	-93

Observations: Customer sign-ups fluctuate throughout the year. The numbers start steady at 110 in January and February, then rise to a peak of 150 in November. There are noticeable dips in May, June, and especially December, which sees a sharp drop to 57 sign-ups. Overall, sign-ups show a mix of growth and decline month-to-month, with a few months standing out as particularly high or low.